



Multivariate statistical analysis of hydrochemical and radiological data of Serbian spa waters

Irena Tanasković^a, Dusan Golobocanin^b, Nada Miljević^{c,*}

^a Institute of Occupational and Radiological Health "Dr Dragomir Karajović", Deligradska 29, 11000 Belgrade, Serbia

^b Vinča Institute of Nuclear Sciences, POB 522, 11001 Belgrade, Serbia

^c Jaroslav Černi Institute for Development of Water Resources, Jaroslava Černog 80, 11226 Belgrade, Serbia

ARTICLE INFO

Article history:

Received 12 January 2011

Accepted 28 August 2011

Available online 10 September 2011

Keywords:

Spa water

PCA

HCA

Natural radioactivity

Serbia

ABSTRACT

Principal component analysis (PCA) and hierarchical cluster analysis (HCA) have been applied in order to recognize and classify spa waters collected at 30 sites on Serbian territory. The data set comprised 30 observations and 18 measured variables; natural radionuclides (^{40}K , ^{238}U , ^{226}Ra , ^{228}Ra), gross alpha (GA) and gross beta (GB) activities, and hydrochemical parameters (temperature, pH, electrical conductivity, total solids, HCO_3^- , Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , SiO_2). A Box–Cox transformation was used as a data pretreatment before the statistical methods applied. Analyzed waters were qualified in 14 categories strongly predominated by hydrogen carbonate. The exploration of the correlation matrix allowed to uncover strong associations between some variables (the alkaline and alkaline-earth elements, the radium isotopes, GA, GB, and chloride) as well as a lack of association between the others (pH, T, Mg^{2+} and SiO_2).

PCA has revealed four latent factors which are responsible for the data structure covering 74.2% of the observed variations among the variables studied. Two of them can be initially assigned to mineralization of the components of the host rock whereas the other PCs are built from variables indicative of natural radioactivity. A reliable grouping of given data set of spa water samples with respect to their geotectonic units was found. The total correct classification of 83.3% was achieved for predefined geotectonic units. The resulting dendrogram of HCA was interpreted to have classified the 30 spa water samples into four major groups and eleven subgroups using 18 variables.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

Numerous mineral, thermal and thermo-mineral water springs are located on Serbian territory (88,000 km²). Geologically, this territory includes a great number of rock complexes (magmatic, sedimentary and metamorphic rocks), markedly different in respect to age, genesis, mineral content and petrochemical and geochemical characteristics (Dimitrijević, 1995). Diversity and complexity of geological structure of the Serbian territory are predetermined by the existence of main geological structure of the Balkan Peninsula which reflects the complexity of the tectonic evolution of the Mediterranean region. Six regional geotectonic units (Pannonian Basin (A), Interior Dinarides (B), Vardar zone (C), Serbo-Macedonian massif (D), Carpatho-Balkans (E), and Dacian basin (F) (Fig. 1)) are significantly different according to the quantity and quality of ground water (Dimitrijević, 1994). The Pannonian Basin extends in the northern part of Serbia with a highly diverse basement consisting of crystalline schists and granitoids in which waters are

predominantly accumulated within the Neogene sediments. The Interior Dinarides and the Carpatho-Balkans areas expand in western and east part of Serbia, respectively. Those units are characterized by the thick deposits of karstified Triassic limestones and non-permeable rocks (Petrović et al., 2010). The Serbian–Macedonian massif located in the central part of Serbia is a crystalline core consisting of two complexes of crystalline schists: lower, highly metamorphosed and pre-Cambrian in age, while the upper complex of green schists ranging in age from Proterozoic to Lower Paleozoic. The Vardar Zone is the most complicated belt of the Balkan Peninsula composed of several blocks of diverse composition, geological history and provenience, and includes characteristic oceanic elements. Outstanding features are rocks of fracture porosity rich in ultramafites, crystalline schists, granitoids and volcanic rocks while metamorphism varies on age. In the central part of Serbia, waters are accumulated within different rocks of a metamorphic complex. Environmental isotope investigations of Serbian spa waters indicated their meteoric origin (Miljević et al., 1996) and age from young (1350 years, the Prolom Spa, no.14) to old water (40,000 years, the Bujanovacka spa, no. 30) (Miliivojević, 1989).

The chemical composition of groundwater is a function of the chemical composition of the aquifer host rocks, flow conditions and residence time in the aquifer (Back and Hanshaw, 1971). Quality of those mineral

* Corresponding author. Tel.: +381 11 390 8388.

E-mail address: emiljevi@vinca.rs (N. Miljević).

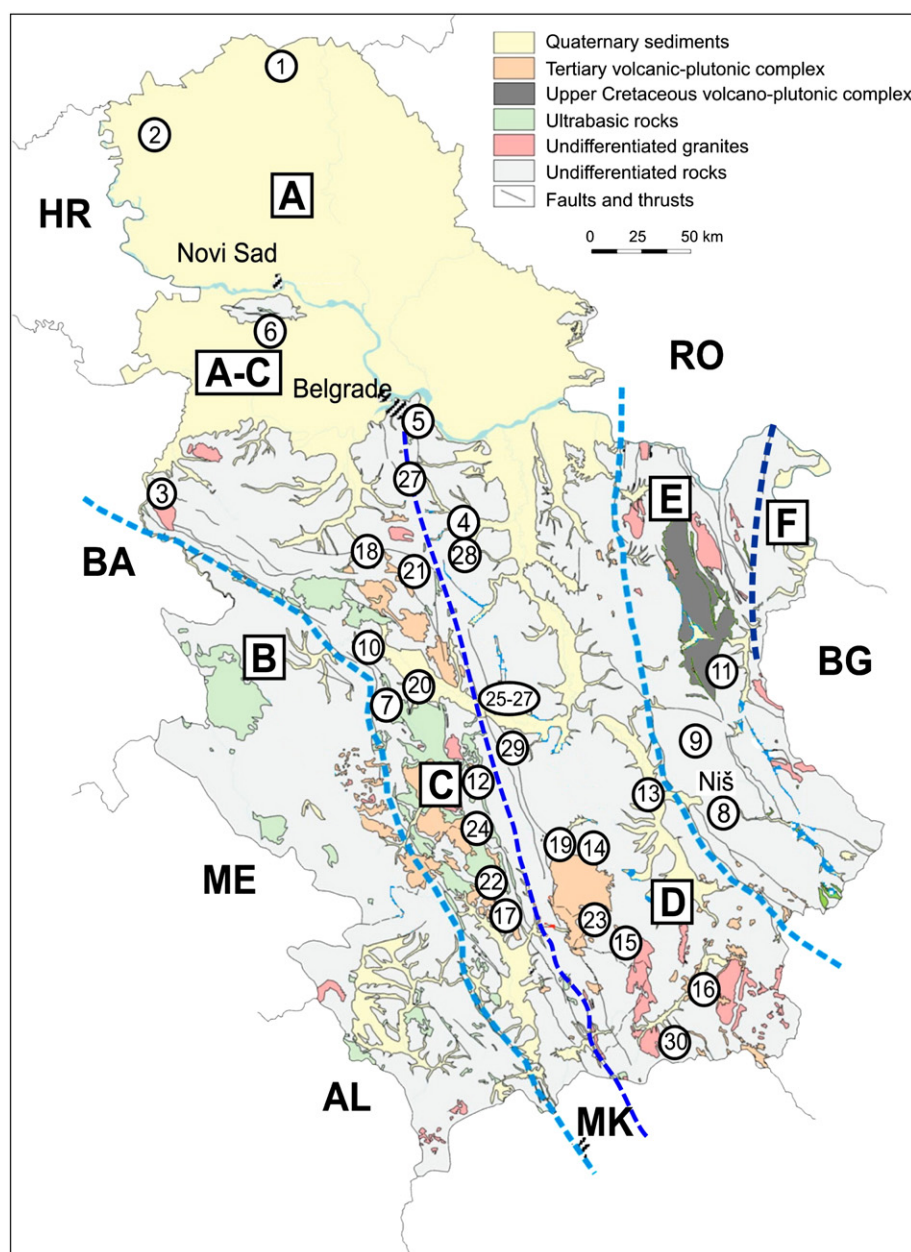


Fig. 1. Map of geotectonic zoning of the territory of Serbia (A. Pannonian Basin; B. Interior Dinarides; C. Vardar zone; D. Serbo-Macedonian massif; E. Carpatho-Balkans; F. Dacian basin) with location of the investigated spa waters.

waters is high and they are used for balneology and medical purposes due to a wide range of soluble matters (CO_2 , H_2S , Se, J, Br, etc.) which have health-improvement effects on human organism. Natural radioactivity in groundwater is directly related to the kind of geological layers crossed by these waters and of their physico-chemical conditions.

Many naturally occurring elements have radioactive isotopes, only potassium (K), and the uranium (U) and thorium (Th) decay series, have radioisotopes that produce gamma rays of sufficient energy and intensity to be measured by gamma ray spectrometry. Average crustal abundances of these elements are in the range 2–2.5% K, 2–3 ppm U and 8–12 ppm Th (IAEA, International Atomic Energy Agency, 2003). ^{40}K with half-life ($t_{1/2}$) of 1.3×10^9 years occurs as 0.012% of natural potassium. Uranium occurs naturally as the radioisotopes ^{238}U ($t_{1/2} = 4.46 \times 10^9$ years) and ^{235}U ($t_{1/2} = 7.13 \times 10^8$ years). Thorium occurs naturally as the radioisotope ^{232}Th ($t_{1/2} = 1.39 \times 10^{10}$ years). Neither ^{238}U , nor ^{232}Th emit gamma rays, and gamma ray emissions from their radioactive daughter products are used to estimate their concentrations. Radium exists

naturally as four isotopes (^{223}Ra , ^{224}Ra , ^{226}Ra , and ^{228}Ra), although only the latter two are of primary concern because of their potential health risks; longer half lives (^{226}Ra ($t_{1/2} = 1600$ years) and ^{228}Ra ($t_{1/2} = 5.57$ years)) and greater likelihood of occurrence.

Chemometric data analysis methods provide powerful tools for the analysis and interpretation of large, environmental, multivariate data sets (Wenning and Erickson, 1994). The multivariate analysis permits the characterization of the spatial variability of natural inhomogeneities, complex problems of interactions and the common appearance of different parameters. Factor analysis, a multivariate statistical method, generates the general relationship between measured variables by showing multivariate patterns that may help to classify and explain the original data. The geological interpretation of factors yields insight into the main processes, which may govern the distribution of hydrochemical variables (Liu et al., 2003).

The principal component analysis (PCA) is multivariate statistical technique used to identify important components or factors that

explain most of the variances of the original system. It is designed to reduce a set of original variables (e.g. groundwater chemical composition) to a small number of indices (i.e., principal components, PCs or factors) for analyzing similarities and differences present among the observed variables that are not readily evident from simple correlation analysis. In principle, PCA (Factor analysis, 1995) approaches produce linear combinations of the original analyzed parameters in a data matrix. These linear combinations represent a type of abstract measurements or factors that are better descriptors for the data-set structure than the original measurements. The minimum requirement to provide a stable solution in PCA is that the number of observations must be greater than the number of independent variables (Yu et al., 1998).

The hierarchical cluster analysis (HCA) is a useful semi-quantitative technique to classify individuals or cases based on their similarities into relatively homogeneous groups called clusters and emphasize their genetic relations. Individuals in each cluster tend to be similar to each other and dissimilar to individuals in the other clusters (Murtagh and Heck, 1987). The most similar objects are first grouped, and these initial groups are merged according to their similarities. The main parameter is the distance between entries relative to the largest distance between any two individual variables. Domain knowledge must be used to guide the formulation of a suitable distance measure for each particular application.

Multivariate statistical analysis has been successfully applied in a number of hydrogeochemical studies to establish the hydrodynamic function of the studied systems (Cloutier et al., 2008), relationship between hydrochemical properties and geological origins of groundwater (Birke et al., 2010; Jang, 2010) but natural radioactivity has rarely been considered (Andreo and Carrasco, 1999; Ramasamy et al., 2011). The objective of this study was to apply multivariate statistics to study relationships among hydrochemical and natural radioactivity parameters of Serbian spa waters. This study gives a contribution to the radionuclides mapping of natural mineral waters. The combination of these research methods should permit a better understanding of hydrogeochemical processes controlling the chemical and radioactive content in aquifer.

2. Material and methods

2.1. Sampling and measurements

Water samples ($n=30$) (Table 1, Fig. 1) were collected from 4 geological structures (hydrogeological basins, karstic terrains, volcanogenic massifs and metamorphic regions) (Protić, 1995) directly on the spot of active therapeutic medical facilities in polyethylene bottles during the period 2007–2008. The samples were transported to the laboratories where gross alpha (GA) and beta (GB) activities, gamma spectrometry analysis and concentrations of principal cations and anions were determined.

The GA and GB activities of water samples and gamma spectrometry measurements of natural radionuclides (^{238}U , ^{228}Ra , ^{226}Ra , ^{40}K) have been carried out in the Institute of Occupational and Radiological Health “Dr Dragomir Karajović” and applied methods have been already described in our previous paper (Tanasković et al., 2011). The outflowing temperature (T), pH, and electrical conductivity (EC) of water were measured immediately after collection on the spot, while major ions (HCO_3^- , Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} and SiO_2) which mainly originate from dissolution were analyzed in the laboratories of Institute of Public Health of Serbia “Dr Milan Jovanovic Batut”, Belgrade. Titration methods were applied to measure alkalinity, Ca^{2+} , and Mg^{2+} . Concentrations of Na^+ and K^+ were determined by atomic absorption spectrophotometry. Ion chromatography is used for the determination of chloride and sulfate ions.

2.2. Multivariate statistical analysis

Statistical treatment of the “raw” (analytical) data presented in this paper was done using the free statistic software package Unscrambler 9.7. Relationships between concentrations of different ions were investigated with factor analyses, which allow the identification of a small number of factors that could explain most of the original data variability (Hopke, 1985). All data were examined using the Shapiro–Wilk’s test

Table 1
Hydrochemical data for the mineral and thermal waters studied on Serbian territory.

No.	Name	T [°C]	pH	EC [μS/cm]	TS [g/L]	Ca ²⁺ [mg/L]	Mg ²⁺	Na ⁺	K ⁺	Cl [−]	SO ₄ ^{2−}	HCO ₃ [−]	SiO ₂	Geological structure
1	Kanjiža	45.0	7.7	3770	4.173	5.6	3.8	1157.0	7.9	56.8	2.0	2903.6	36.7	Hydrogeological basin
2	Junaković	25.0	7.3	5100	6.200	24.0	24.0	476.0	4.0	369.0	2.0	829.0	25.0	
3	Koviljača	25.0	6.6	677	0.641	120.0	36.6	200.0	22.2	89.0	55.0	890.0	18.0	
4	Selters	48.0	7.0	2219	2.119	60.0	18.0	2280.0	30.0	1560.0	2.0	3360.0	20.0	
5	Višnjička	14.4	7.5	1100	1.100	56.0	48.8	226.1	3.5	198.8	82.0	573.4	22.0	Karstic terrains
6	Vrdnik	33.0	7.4	722	0.538	60.0	67.0	69.0	7.1	28.0	90.0	555.0	10.0	
7	Bogutovačka	25.0	7.2	440	0.377	50.0	61.0	28.0	1.5	10.0	15.0	440.0	20.0	
8	Niška	39.1	7.6	265	0.313	62.0	14.0	22.7	1.2	28.0	40.0	220.0	18.0	
9	Soko Park	43.0	7.5	248	0.200	74.0	17.0	7.0	8.0	14.0	20.0	295.0	18.0	Volcanogenic massifs
10	Ovčar	35.0	7.0	324	0.322	102.0	32.0	13.0	2.5	14.0	18.0	456.0	25.0	
11	Gamzigradska	34.0	7.9	531	0.460	60.0	23.0	149.0	3.2	86.0	28.0	270.0	35.0	
12	Jošanica	26.1	7.3	350	0.322	50.0	25.0	6.4	3.4	7.0	30.0	256.0	22.0	
13	Ribarska	38.0	9.0	413	0.489	4.0	0.2	92.0	1.7	7.0	53.0	122.0	60.0	
14	Prolom	26.4	9.0	211	0.228	4.0	0.1	48.4	0.2	7.0	2.0	79.3	60.0	
15	Sijarinska	45.0	7.5	2040	1.900	50.0	30.5	1150.0	46.8	92.0	96.0	3233.0	40.0	
16	Vranjska	95.0	6.9	1256	0.665	30.0	2.4	326.0	8.6	57.0	365.0	384.3	90.0	
17	Banjska	40.0	6.7	640	0.630	31.0	28.5	300.0	14.4	75.0	65.0	770.0	52.0	
18	Ljig	30.0	7.2	917	1.180	40.0	18.3	276.4	13.2	35.5	5.0	902.8	19.3	
19	Kuršumliska	53.0	7.1	2550	3.168	26.0	33.0	713.0	34.0	28.0	14.0	2135.0	120.0	
20	Mataruška	30.0	6.9	1263	1.422	50.0	58.0	230.0	9.7	63.0	27.0	915.0	40.0	
21	Gornja Trepča	31.0	7.5	395	0.362	26.0	61.0	10.0	1.3	10.0	8.0	385.0	18.0	Metamorphic regions
22	Vuča	30.0	8.4	110	0.100	48.0	0.5	15.6	0.6	17.0	0.9	164	2.0	
23	Tularska	21.7	7.4	3703	2.3364	178.0	74.0	1090.0	31.0	147.0	486.0	2684.0	36.0	
24	Novopazarska	40.0	6.8	1700	1.600	35.0	27.0	378.0	10.5	42.0	16.0	1170.0	45.0	
25	Vrnjačka Jezero	25.5	6.7	2658	3.120	120.0	97.0	447.0	43.0	31.0	6.0	2040.0	45.0	
26	Vrnjačka Snežnik	17.0	6.7	1380	1.496	122.0	95.0	447.0	43.0	31.0	6.0	2040.0	45.0	
27	Vrnjačka Topla	36.0	6.5	2480	2.758	123.0	100.0	447.0	43.0	31.0	6.0	2040.0	45.0	
28	Bukovička	20.0	7.5	1100	0.990	100.0	24.0	1170.0	43.0	21.0	2.0	3538.0	110.0	
29	Veluce	21.0	6.5	1800	1.750	80.0	293.0	195.0	12.0	39.0	4.0	2171.0	35.0	
30	Bujanovačka	30.0	6.8	2650	2.300	40.0	21.0	1443.0	47.0	67.0	180.0	3086.0	30.0	

(Shapiro and Wilk, 1965) for normal distribution choosing $p < 0.05$, which indicated a significant departure from normality. Where data failed on the Shapiro–Wilk's test, a Box–Cox (Box and Cox, 1964) transformation was performed to obtain the standard normal distribution data. The Box–Cox transformed data which passed the normality test ($p > 0.05$) together with non transformed data were used for chemometric analysis (correlation analysis, cluster analysis, PCA).

In summary, PCA decomposes the original matrix consisting of hydrochemical and radioactive data taken as variables (18 columns) and sampling spa sites as mathematical–statistical cases taken as observations (30 rows) into several products of multiplication into loading and score matrices. The number of PCs extracted from the variables was determined by a scree test according to Kaiser's rule (Kaiser and Rice, 1974). This criterion retains only factors with eigenvalues that exceed one. In addition, we applied Bartlett's test (Jackson 1993) for the significance of the PCA residuals, which is comparable to a chi-square distribution. It tests equality for a set of successive eigenvalues derived from covariance matrix. It is used a sequential procedure; at first testing that the eigenvalues of all components are equal. The null hypothesis is that the eigenvalues of components not retained equal. If this is rejected, we then test equality of eigenvalues of all components except the first, and so on. Once we do not reject the null hypothesis, we retain all components above those being tested. In our case, the first four components were statistically significant at an alpha level of 0.05 for $n = 30$ and degrees of freedom ($df = 87$). For this study, a factor correlation coefficient greater than 0.70 or 70% was considered significant. Our intention was to perform data processing in the same way for all cases preferring such a way of the factor rotation to get as many positive loadings as possible in order to achieve a more meaningful and interpretable solution. The Varimax rotation (orthogonal) was applied to 'clean up' the PCs by increasing the participation of the variables with higher contribution.

The hierarchical cluster analysis (HCA) with completed linkage was applied to identify clusters of 30 individual spa waters with 18 variables consisting of the main hydrochemical (temperature, pH, EC, TS, HCO_3^- , Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , SiO_2) and natural radioactivity data (GA, GB, ^{40}K , ^{238}U , ^{226}Ra , ^{228}Ra). The result of such classification is visualized as a dendrogram in which the dissimilarity of the objects (systems) is represented by the height at which the branches are connected. The complete linkage of Euclidean distance (straight line distance between two points in c -dimensional space defined by c variables) is used, i.e. the maximum dissimilarity between the objects in the clusters is considered.

3. Results and discussion

3.1. Hydrochemical and radiological data

Mineral waters originating from Serbian territory demonstrate a wide range of temperature (T) from cold (14.4°C , no. 5) to very hot springs (95°C , no. 16). A wide variation in pH (6.5–9.0), electrical conductivity (EC, 110–5100 $\mu\text{S}/\text{cm}$), and total solid content (TS, 110–6200 mg/L) reflect the compositional difference between them (Table 1). The spa waters (nos. 14, 25–30) are exploited as drinking water, while the remaining springs are used as popular spas for curative thermal purposes.

The chemical content of analyzed water samples are presented with a Piper trilinear diagram (Piper, 1944) (Fig. 2). In order to present the composition of water as precisely as possible, cations or anions are listed in the decreasing order starting with the most prevailing one. With this procedure, the majority of the waters analyzed are Na-HCO_3 (nos. 1, 14–15, 17–19, 23–24, 28, 30), Na-Mg-HCO_3 (nos. 20, 25–27), $\text{Na-HCO}_3\text{-Cl}$ (nos. 2, 4), $\text{Na-HCO}_3\text{-SO}_4$ (nos. 13, 16), Ca-Mg-HCO_3 (nos. 8–10) types. Such grouping also agrees well with the clustering shown in the Piper diagram. The other waters are defined as Na-Ca-HCO_3 (no. 3), $\text{Na-Ca-HCO}_3\text{-Cl}$ (no. 11), $\text{Na-Mg-HCO}_3\text{-Cl}$ (no. 5), Ca-HCO_3 (no. 12), Ca-Na-HCO_3 (no. 22), Mg-HCO_3 (no. 21),

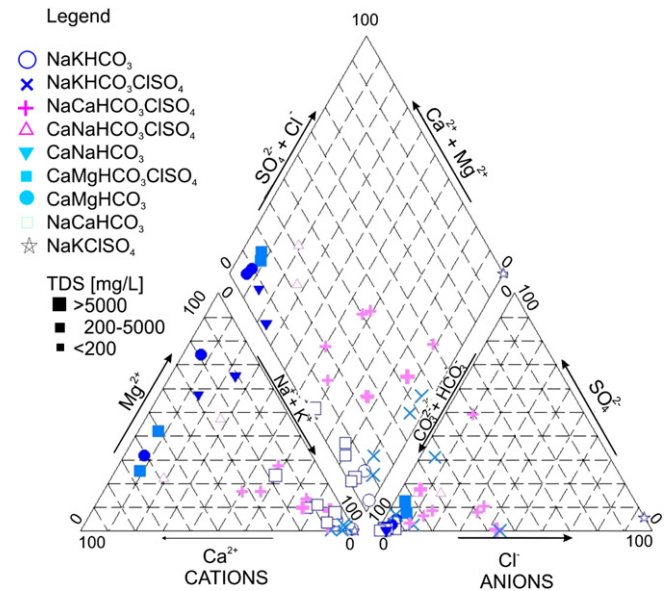


Fig. 2. Piper-tri-linear diagram of the Serbian spa waters based on water type.

Mg-Ca-HCO_3 (no. 7), Mg-Na-HCO_3 (no. 29), Mg-Ca-Na-HCO_3 (no. 6) according to 143 different types of European mineral waters (Gros, 2003). HCO_3^- and Na^+ dominant water normally indicates ion-exchanged waters, although the generation of CO_2 at depth can produce HCO_3^- where Na^+ is dominant under certain circumstances (Chadha, 1999). Groundwater in limestone, sandstone, and many other aquifers in recharge areas is dominated by HCO_3^- and Ca^{2+} .

For further classification of the analyzed waters, we plotted a Van Wierdum's diagram (van Wierdum, 1980), the ionic ratio (IR) vs. the electric conductivity (EC), where $\text{IR} = [\text{Ca}^{2+}] / ([\text{Ca}^{2+}] + [\text{Cl}^-])$ with concentrations expressed in meq/L (Fig. 3). The EC can be seen as a measure of the salinity, the IR as a measure of the prevalence of calcium among the cations and chloride among the anions. This prevalence is a result of geochemical processes that occurred within the aquifer. Based on the position in the Van Wierdum's diagram, the groundwater can be classified as atmospheric, At (rainwater), lithotrophic, Li (calcium-rich freshwater), thalassotrophic, Th (seawater) or anything in between. The samples are distributed among the three reference water types from the hydrological cycle. Most of samples appear to be predominantly in the zone of

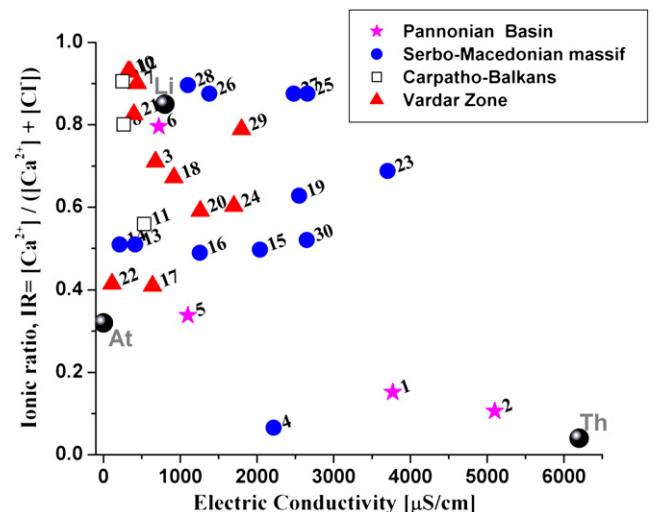


Fig. 3. Van Wierdum's diagram for Serbian spa waters based on geotectonic zoning location. (At – atmospheric, Li – lithotrophic, Th – thalassotrophic groundwater).

lithotrophic water. However, a few of the samples (nos. 1, 2) from the Pannonian Basin with higher chloride content have a composition similar to that of seawater as the consequence of the ancient Pannonian Sea.

The GA activities measured in 30 spa waters span a range of 2 orders of magnitude, from <10 mBq/L to 470 mBq/L while GB activities ranged from 108 to 2206 mBq/L. Activity concentration of radium was generally low (10–530 mBq/L for ^{226}Ra and 10 to 680 mBq/L for ^{228}Ra). The highest concentrations of both ^{226}Ra (nos. 4, 8, 28) and ^{228}Ra (nos. 15, 23, 25) were measured in the spa waters emerging from limestone and marble as the dominant geological structure, respectively. The latter occurred in the zone of carbogaseous spring where the different water types originate from mixing of groundwater and highly mineralized carbogaseous fluids from depth greater than 2000 m (Milivojević, 2003). The ^{226}Ra is present in all rocks and soils in variable amounts and originates from the radioactive series of ^{238}U while its high values are usually associated with deep sandstone aquifers (Iyengar, 1990).

Despite of a non-constant $^{228}\text{Ra}/^{226}\text{Ra}$ activity concentration ratio, it seems possible to conclude that groundwater with high ^{226}Ra content contains less ^{228}Ra content. The highest ratio of ^{226}Ra to ^{228}Ra activity reaching 13.2 in Niska spa (no. 8) can be related to a specific type of limestone convective systems built up of fractured Jurassic and Cretaceous limestones, open on two sides for feeding up with meteoric water and according to hydrogeological classification it is designated as “thermo-karstic spring” (Milivojević, 1991). The high radium contents in the soil (over 800 Bq/kg for ^{226}Ra) and the groundwater in this area may be due to the presence of uranium mineralization (Žunić et al., 2006). ^{238}U activity concentrations were in the range of 40–380 mBq/L, with a mean value of 110 ± 60 mBq/L, and the most remarkable is in the Tularska spa (no. 23). From the ^{238}U activity concentration measured, the U natural level is calculated, assuming a radioactive equilibrium of the uranium isotopes and the activity of 25.28 mBq from 1 μg of natural uranium (i.e. 1 μg of ^{238}U isotope produces 12 mBq) (Kitto et al., 2005). Therefore, the weight concentration of total uranium in analyzed water would be in the range from 3 $\mu\text{g}/\text{L}$ (no. 13, Na– HCO_3 – SO_4 type) to 32 $\mu\text{g}/\text{L}$ (no. 23, Na– HCO_3 type) that is in line with reported values in Serbia up to 41 $\mu\text{g}/\text{L}$ in eastern part (Nikic et al., 2008) and 2700 $\mu\text{g}/\text{L}$ in abandoned mines waters (Dragisic et al., 2008).

3.2. Multivariate statistical analysis

3.2.1. Correlation coefficient analysis

The individual radionuclide concentration in water can be affected by a number of factors, such as (1) the geological compositions of the

soil or rocks, (2) the mineral contents in water, (3) the chemical behavior of the nuclides or elements (Jia and Torri, 2007). Pearson correlation coefficients between hydrochemical and radiological parameters measured in spa waters were calculated and in the form of correlation matrix summarized in Table 2. The terms ‘strong’, ‘moderate’, and ‘weak’ as applied to correlation coefficients, refer to absolute coefficient values of >0.75, 0.75–0.50 and 0.50–0.36, respectively. Alpha level for testing significance of correlations was maintained at $p < 0.05$ for samples ($n = 30$).

We observe strong and positive correlations of hydrogencarbonate with Na^+ and K^+ ($r = 0.85$, 0.88 respectively), and Na^+ with K^+ , Cl^- , ^{40}K and GB ($r = 0.78$, 0.71 , 0.73 and 0.70 , respectively). Most of these associations can be explained in terms of the hydrochemical characteristics of the aquifer. The EC is highly associated to HCO_3^- and TS ($r = 0.83$, 0.98 respectively) because TS includes both dissolved and suspended solids, which was determined during preparation of each sample for the GA measurement (Kitto et al., 2005). The highest positive correlations appear between GB activity and ^{40}K activity and K^+ concentration ($r = 0.92$, 0.79 respectively) due to the GB activity in ground water is primarily composed of ^{40}K (50–100%) and uranium decay products. Such a result would be in line with the findings that total radioactivity increases with the total mineral content (Ben Fredj et al., 2005) because a higher natural radioactivity is mainly related to the higher mineralization in waters of underground origin.

Moderate positive correlations were resolved for Ca^{2+} with Mg^{2+} , ^{238}U , GB and ^{228}Ra ($r = 0.64$, 0.55 , 0.55 and 0.53 respectively). The ^{226}Ra activity is very weak related to the activity of ^{228}Ra ($r = 0.30$) because its origins are in different decay series, in which the parent ^{232}Th and ^{238}U have quite dissimilar chemical behaviors in aquifers. The contribution of each radionuclide to the total radioactivity varies considerably across the sampling sites. Moreover, the $^{226}\text{Ra}/^{228}\text{Ra}$ ratio at various sites ranged between 0.13 and 5.7 and only about 7% of the sites displayed values around 1.0 (0.8–1.2). This implies that the concentration of ^{226}Ra in groundwater is not related to uranium dissolved.

3.2.2. Principal components analysis (PCA)

Principal component analysis (PCA) was carried out to identify the underlying pattern between the measured hydrochemical and radioactive parameters and to help the data interpretation. The input data set consisted of a 30×18 matrix, in which rows represented spa water samples analyzed (30 objects), and columns the specific analyzed parameters (variables); pH, T, EC, principal ions (HCO_3^- , Ca^{2+} ,

Table 2

Pearson's correlation matrix for hydrochemical parameters and natural activity measured in Serbian spa waters at the probability of 95% for each sampling site.

	pH	T	EC	GA	GB	^{40}K	^{226}Ra	^{228}Ra	^{238}U	TS	HCO_3^-	Cl^-	SO_4^{2-}	Na^+	K^+	Ca^{2+}	Mg^{2+}	SiO_2
pH	1																	
T	–0.01	1																
EC	–0.49	0.01	1															
GA	–0.20	0.06	0.32	1														
GB	–0.55	–0.05	0.72	0.50	1													
^{40}K	–0.42	–0.04	0.67	0.41	0.92	1												
^{226}Ra	0.05	–0.01	–0.04	0.63	0.38	0.23	1											
^{228}Ra	–0.09	–0.19	0.43	0.13	0.59	0.53	0.30	1										
^{238}U	–0.05	–0.15	0.14	–0.13	0.37	0.24	0.05	0.60	1									
TS	–0.47	–0.03	0.98	0.38	0.69	0.65	–0.04	0.36	0.05	1								
HCO_3^-	–0.59	–0.08	0.83	0.37	0.79	0.78	0.12	0.53	0.09	0.81	1							
Cl^-	–0.42	0.04	0.67	0.25	0.54	0.40	0.17	0.31	0.14	0.63	0.57	1						
SO_4^{2-}	–0.11	0.20	0.01	–0.32	0.10	0.02	–0.10	0.16	0.28	–0.1	–0.09	0.14	1					
Na^+	–0.33	0.07	0.85	0.28	0.70	0.73	0.08	0.49	0.07	0.83	0.85	0.71	0.00	1				
K^+	–0.63	0.01	0.69	0.32	0.79	0.81	0.06	0.52	0.18	0.67	0.88	0.48	0.10	0.78	1			
Ca^{2+}	–0.54	–0.39	0.15	0.21	0.55	0.40	0.25	0.53	0.55	0.1	0.39	0.24	0.14	0.10	0.49	1		
Mg^{2+}	–0.70	–0.47	0.36	0.25	0.43	0.31	–0.04	0.26	0.17	0.36	0.47	0.20	0.05	0.11	0.43	0.64	1	
SiO_2	–0.11	0.22	0.39	0.24	0.30	0.38	–0.05	0.16	–0.07	0.39	0.32	–0.01	0.08	0.47	0.41	–0.14	–0.01	1

‘Strong’ correlation coefficient >0.75 (bold, statistical significant at the 0.05 level).

‘Moderate’ correlation coefficient 0.75–0.50.

‘Weak’ correlation coefficient 0.50–0.36.

Table 3

Loadings of 18 hydrochemical and radiological variables on four significant principal components for Serbian spa waters.

Factor Loadings (Varimax normalized) (BoxCox0.sta) Extraction: Principal components (Marked loadings are >.700000)					
Parameter	Abbreviation	PC1 (37.5%)	PC2 (13.8%)	PC3 (10.5%)	PC4 (12.4%)
pH	pH	−0.558894	−0.599592	0.140810	0.041739
Temperature	T	0.129825	−0.666656	−0.072136	−0.043697
Elc. conductivity	EC	0.932255	0.067803	−0.023200	0.020184
gross alpha	GA	0.354519	0.097436	0.791293	−0.225046
gross beta	GB	0.771735	0.227345	0.357833	0.359443
⁴⁰ K	K4	0.776664	0.109612	0.290649	0.268652
²²⁶ Ra	R6	−0.011695	−0.034704	0.900503	0.180492
²²⁸ Ra	R8	0.399390	0.104150	0.258765	0.714837
²³⁸ U	U8	0.030841	0.176562	−0.009444	0.843785
Total solids	TS	0.921497	0.087648	0.018808	−0.099182
HCO ₃ [−]	HC	0.893782	0.251005	0.141157	0.063790
Cl [−]	Cl	0.654913	0.067603	0.056037	0.136754
SO ₄ ^{2−}	SO	0.029585	−0.126665	−0.418986	0.585962
Na ⁺	Na	0.928129	−0.143229	0.077568	0.098041
K ⁺	K	0.845670	0.222414	0.066618	0.217205
Ca ²⁺	Ca	0.161498	0.723589	0.215700	0.530998
Mg ²⁺	Mg	0.314195	0.860436	−0.031869	0.029250
SiO ₂	Si	0.530142	−0.356258	−0.013413	−0.044943
Expl.Var		6.743603	2.477092	1.999335	2.229205
Prp.Tot		0.374645	0.137616	0.111074	0.123845

Mg²⁺, Na⁺, K⁺, Cl[−], SO₄^{2−}, SiO₂, and radionuclides (GA, GB, ⁴⁰K, ²³⁸U, ²²⁶Ra, ²²⁸Ra).

The PCA reduced the parameters to four latent factors explaining 74.2% of the sampling variance responsible for the data structure; the rotated factor loadings for each parameter are presented in Table 3. Loading values, that represent the importance of the variables for the components (i.e., correlation coefficients >0.7), are marked in boldface type. The first component (PC1) was positively correlated with EC (0.93), TS (0.92), Na⁺ (0.93), HCO₃[−] (0.89), K⁺ (0.84), ⁴⁰K (0.78), and GB (0.77) accounted for about 37.5% of the total variance in the data set and could be conditionally named the “mineralization or residence time” factor (Andreo and Carrasco, 1999) because these interactions originating from geological components of aquifer.

The PC2 (13.8%) is determined by with high-positive loadings Mg²⁺ (0.86) and Ca²⁺ (0.72). This component could be conditionally named as the “hardness” component because two ions generally used to calculate hardness (Cloutier et al., 2008) and are typically associated with the dissolution of carbonate rocks in those aquifer systems (Fig. 4).

The PC3 covered 10.5% of the total variance and is clearly characterized by highly positive loadings in ²²⁶Ra (0.90) and GA (0.79) and can be attributed to gross alpha factor. The GA activity in the

groundwater originates from the ²³⁸U and its daughter ²²⁶Ra because the evaporation procedure that was used for measurement excludes radon short-lived progeny, ²²²Rn (Kitto et al., 2005). The last principal component, accounting for about 12.4% of the total variation in the dataset, is dominated by ²³⁸U (0.84) and ²²⁸Ra (0.71) demonstrating the influence of both ²³⁸U and ²³²Th decay series. The moderate inter-correlation between this isotope pair (r = 0.60, Table 2) supports that this factor can be related to the mixed radioactive decay series. The source rocks (metamorphic, magmatic and sedimentary) of different geological age from Proterozoic to Mesozoic contain the Th/U concentration ratio between 2.8 and 4.8 on Serbian territory (Miliivojević, 1989).

In score plot PC1 vs. PC2 (PC2, PC3, and PC4) (Fig. 5) the groundwater sampling sites are divided into three distinct clusters mainly according to their geotectonic units. Observations that belong to the cluster I plotted in the upper left corner of the graphic, contains waters with the longest residence time comparable to an apparent age of 20,000–40,000 years (Miliivojević, 1989) based on radiocarbon dating. They belong to Serbo-Macedonian massif and appear in metamorphic (nos. 25–28, 30) and volcanic (nos. 15, 19, 23) areas.

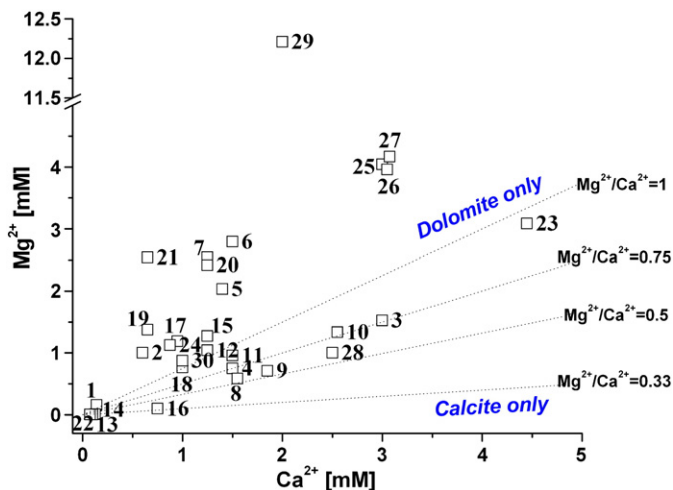


Fig. 4. Mg²⁺ versus Ca²⁺ indicates the dominance of dolomite dissolution in groundwater aquifers.

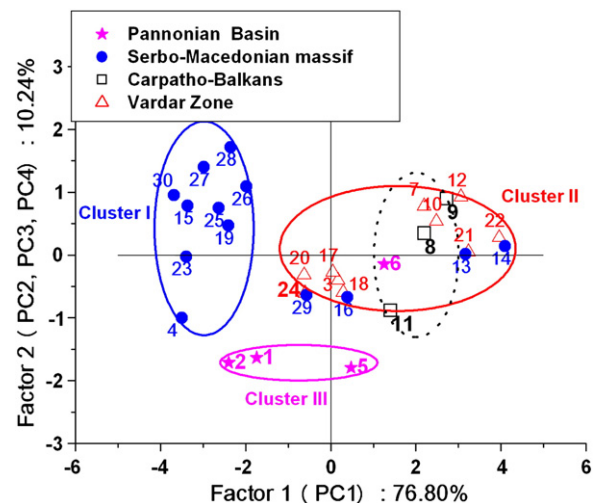


Fig. 5. Score plot of PC1 and PC2 (PC2, PC3, and PC4) illustrating the differentiation of spa samples for all cases (1–30) used in this study according to their geotectonic units. Sampling places designated by numbers in clusters are given in Table 1.

Table 4
Classification matrix.

Class (geotectonic unit)	Cluster I	Cluster II		Cluster III
	Serbo-Macedonian massif (D)	Vardar zone (C)	Carpatho-Balkans (E)	Pannonian Basin (A)
No. of samples	12 (8)	11(11)	3(3)	4 (3)
Corr. class (%)	66.7	100	100	75

Samples from the cluster II are plotted in the right part of the plot, including mainly waters from Vardar zone (nos. 3, 7, 10, 12, 17, 18, 20, 21, 22, 24) draw from volcanogenic massive as well as waters (nos. 13, 14, 16) from Serbo-Macedonian massif. This cluster also contains waters emerging from karstic region in geotectonic unit Carpatho-Balkans (nos. 8, 9, 11). The residence times of those waters are shorter, less than 20,000 years old because most likely water flows faster through a well developed karst drainage system or fissured rocks. The samples (nos. 1, 2, 5) in the cluster III, scattered between cluster I and cluster II, belong to the Pannonian Basin. No doubt, that water from Vrdnik spa (no. 6) is an outlier. According to available data, their age is not known. Nevertheless, the fact that highest activities of ^{226}Ra , ^{228}Ra and ^{238}U have been detected in waters which belong to the cluster I with longest residence times imply that those waters remain in the aquifer long enough allowing prolonged contact with the relevant rocks and dissolution of radioactive elements. The calculated saturation indices ($\text{SI}_{\text{calcite}}$) for all analyzed waters exceed value of +0.05 indicating their supersaturation with respect to calcite. The values gradually decrease with decreasing the apparent age of water and for those in cluster I they are between 0.74 and 1.88, while for those in cluster 2 from 0.3 to 1.04.

Quantitative estimation of the classification efficiency could be obtained from the classification matrix (Table 4), which shows the predicted number of samples in each class according to geotectonic features and percentage of correctly classified samples (in bracket). The classification rate ranged from 66.7 to 100% with an overall correct classification of 83.3%. Waters from Carpatho-Balkans and Vardar zone were classified correctly. Four (nos. 13, 14, 16, 29) spa waters of the Serbo-Macedonian massif were classified incorrectly into the

group of those from the Vardar zone as well one (no. 6) from the Pannonian Basin, indicating rather unique hydrochemical characteristics.

3.2.3. Cluster analysis

HCA is used to classify waters into objective groups. A subjective evaluation was made choosing the horizontal line (phenon line) (Sneath and Sokal, 1973) across a dendrogram at a linkage distance of about 58% (Fig. 6) based on its visual examination to obtain four clusters (main groups) as it was resulted from the PCA. The outcome dendrogram shows four main groups (I–IV) and 8 subgroups of 30 water samples that have similar hydrochemical and radioactive characteristics. The samples clustered under each subgroup and the average values for each of the parameters are presented in Table 5. These values for each of the composition of the subgroups produced by the HCA analysis reveal trends between them and they are the basis for the distinction of the subgroups. Group IV samples all have significantly higher TS than group I, II or III samples which also appear to be separated based on TS. We selected the division into subgroups using a 40% index of dissimilarity, which proceeds clearly distinguishable in terms of their hydrogeological and hydrochemical variables.

Samples that belong to the same subgroup are located in close proximity to one another suggesting more or less the same hydrogeochemical processes (evolution) and/or flow paths. The most subgroups are confined within a particular geotectonic unit (e.g. SB-2 Serbian-Macedonian Massif). On the other hand, the division into subgroups is not always so apparent. For instance, SB-2 and SB-3 appear to be similar in hydrochemical parameters, but they belong to different geotectonic units (Serbo-Macedonian massif and Vardar zone, respectively) as well as 8 cases in the largest SB-5, which are

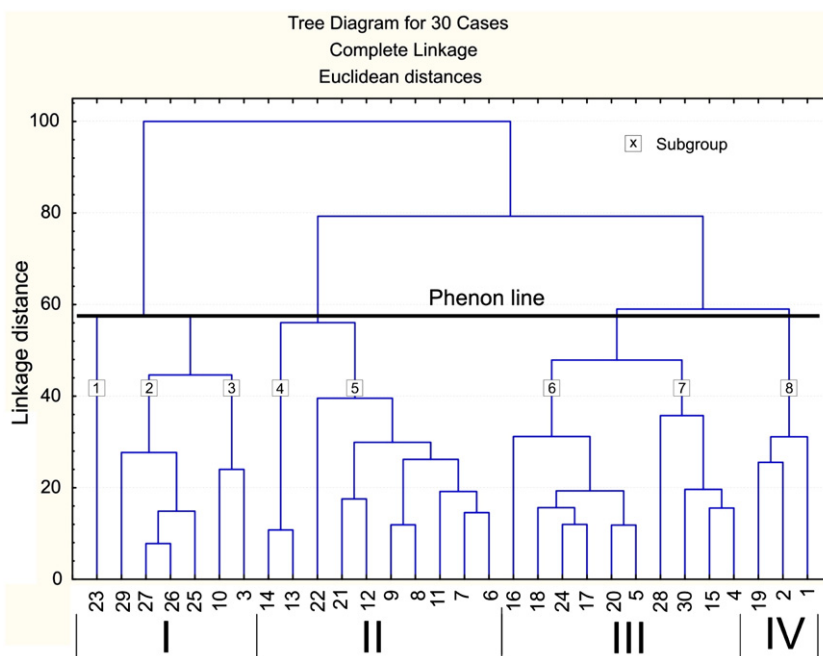


Fig. 6. Hierarchical dendrogram obtained by the site clustering analysis with both hydrochemical and natural radioactivity parameters for 30 spa waters. The notation represents the same spa analyzed as in Table 1.

Table 5

Average values for hydrochemistry and natural radioactivity parameters of the spa water subgroups derived from HCA.

Group	Sub group	n	pH	Tem [°C]	EC [μS/cm]	TS [mg/L]	HCO ₃	Na	K	Ca	Mg	Cl	SO ₄	SiO ₂	GA mBq/L	GB	⁴⁰ K	²²⁶ Ra	²²⁸ Ra	²²⁸ U
I	1	1 (23)	7.4	21.7	3703	2336	2684.0	1090.0	31.0	178.0	74.0	147.0	486.0	36.0	10	2206	1290	100	680	380
	2	4 (25, 26, 27, 29)	6.6	24.9	2080	2281	2072.8	384.0	35.3	110.0	146.0	33.0	5.5	42.5	172	1263	1200	60	140	120
	3	2 (3, 10)	6.8	30.0	501	482	673.0	106.5	12.4	111.0	34.3	51.5	28.0	21.5	19	338	300	90	30	110
II	4	2 (13, 14)	9.0	32.2	312	358	100.7	70.2	0.9	4.0	0.1	7.0	27.5	60.0	15	114	90	20	10	70
	5	8 (6, 7, 8, 9, 11, 12, 21, 22)	7.6	32.7	383	335	323	38.5	3.3	53.8	33.6	25.0	29.0	17.9	106	328	251	148	25	114
III	6	6 (5, 16, 17, 18, 20, 24)	7.0	41.6	1146	1100	785.9	289.4	10.0	40.3	30.5	78.6	93.3	44.7	35	397	410	17	18	90
	7	4 (4, 15, 30, 28)	7.2	35.8	2002	1827	3304.3	1510.8	41.7	62.5	23.4	435.0	70.0	50.0	228	1548	1630	240	220	80
IV	8	3 (1, 2, 19)	7.4	41.0	3807	4514	1955.9	782.0	15.3	18.5	20.3	151.3	6.0	60.6	183	788	840	90	20	90

n—number of samples within subgroup.

located in Vardar zone and Pannonian and Carpatho-Balkans basins. However, SB-1 and SB-2 are emerged in the same geotectonic unit (Serbo-Macedonian massif), but they appear to be distinguished by the values for pH, Na⁺, Cl[−] and GB.

The statistically defined clusters of samples have less physical meaning than PCA (Fig. 5) because HCA uses less information (distances only) than PCA. The good statistical coherence among mean values of the subgroups cannot be clearly explained in terms of the hydrochemical and radioactive variables for characterization of groundwater circulations and surface water-groundwater interactions.

4. Conclusions

Type and chemical content of mineral waters depend upon many geological, lithological and tectonic factors. According to 143 different types of European mineral waters, the analyzed Serbian spa waters are highly enriched in alkaline and alkaline-earth elements and classified into 14 different categories (i.e. Na–HCO₃, Na–Mg–HCO₃, Na–HCO₃–Cl, Na–HCO₃–SO₄, Ca–Mg–HCO₃, Na–Ca–HCO₃, Na–Ca–HCO₃–Cl, Na–Mg–HCO₃–Cl, Ca–HCO₃, Ca–Na–HCO₃, Mg–HCO₃, Mg–Ca–HCO₃, Mg–Na–HCO₃, Mg–Ca–Na–HCO₃) on the basis of the sum total of the cations and anions that contribute at least 20 meq-% to the ionic content. Most of the samples are of the sodium-bicarbonate type.

Of the 18 analyzed hydrochemical (T, pH, EC, HCO₃[−], Ca²⁺, Mg²⁺, Na⁺, K⁺, Cl[−], SO₄^{2−} and SiO₂) and natural radioactivity (GA, GB, ²³⁸U, ²²⁸Ra, ²²⁶Ra, ⁴⁰K) parameters, the highest correlations ($r > 0.78$) are observed between the alkaline, and alkaline-earth elements, the radium isotopes, GA, GB, and Cl[−]. Moderate positive correlation coefficients ($0.53 < r < 0.64$) between Ca²⁺ and ²²⁸Ra, ²³⁸U, GB, and Mg²⁺ was found. There is not any correlation for following parameters: pH, T, Mg²⁺ and SiO₂.

PCA is able to characterize spa waters according to four statistically significant eigenvalues > 0.7 accounting for 74.2% of the total variance. The first component (PC1) explains 37.5% of the variance, and thus, account for the majority of the variance in the original dataset. It is correlated by means of very high coefficients with EC (0.93), TS (0.92), Na⁺ (0.93), HCO₃[−] (0.89), K⁺ (0.84), ⁴⁰K (0.78), and GB (0.77) which can be assigned to 'mineralization' processes originating from the chemical interaction between groundwater and the host rocks. Components 2–4 are not as important, and each of these three components explains about the same amount of variance, between 10.5% and 13.8%. The second component (PC2) is associated with the dissolution of carbonate rocks while PC3 and PC4 are correlated with natural radioactivity reflecting the influence of both ²³⁸U and ²³²Th decay series. Using the HCA analysis to initially cluster the samples (into groups and subgroups) provides an efficient means to recognize groups of samples that have similar hydrochemical and radioactive characteristics.

The results of this study demonstrate the importance of the residence times of groundwater, i.e. the average age of the water in the reservoir which is strongly correlated to geotectonic unit. The most

obvious reason for discrimination of the samples seen in the PCA biplot was the mode of mobilization of groundwater and chemical/physical processes involved in the transport of these nuclides in groundwater systems and complexity of the processes involved in their accumulation in groundwater. Naturally, all of these statements are valid within the scope of this study and the samples chosen to be analyzed here.

Acknowledgments

This work was supported by the Ministry of Science and Technological Development of the Republic of Serbia under project number III 43009. We thank Prof. S.M. Komatina Petrović for many fruitful discussions.

References

- Andreo, B., Carrasco, F., 1999. Application of geochemistry and radioactivity in the hydrogeological investigation of carbonate aquifers (Sierras Blanca and Mijas, southern Spain). *Applied Geochemistry* 14, 283–299.
- Back, W., Hanshaw, B.B., 1971. Geochemical interpretations of groundwater flow systems. *W.C. Bull.* 7, 1008–1016.
- Ben Fredj, A., Hizem, N., Chelbi, M., Ghedira, L., 2005. Quantitative analysis of gamma-ray emitters radioisotopes in commercialised bottled water in Tunisia. *Radiation Protection Dosimetry* 117, 419–424.
- Birke, M., Rauch, U., Harazim, B., Lorenz, H., Glatte, W., 2010. Main and trace elements in German bottled water, their regional distribution, and accordance with national and international standards. *Journal of Geochemical Exploration* 107, 245–271.
- Box, G.E.P., Cox, D.R., 1964. An analysis of transformations. *J. R. Stat. Soc. B* 26, 211–252.
- Chadha, D.K., 1999. A proposed new diagram for geochemical classification of natural waters and interpretation of chemical data. *Hydrogeology Journal* 7, 431–439.
- Cloutier, V., Lefebvre, R., Therrien, R., Savard, M.M., 2008. Multivariate statistical analysis of geochemical data as indicative of the hydrogeochemical evolution of groundwater in a sedimentary rock aquifer system. *Journal of Hydrology* 353, 294–313.
- Dimitrijević, M., 1995. *Geology of Yugoslavia*. Geoinstitut, Belgrade (in Serbian).
- Dimitrijević, M., 1994. Geological map 1:2 000 000. Geological atlas of Serbia, no 1.
- Dragisic, D.V., Savic, N., Vranjes, A., 2008. Environmental impact of uranium mine waters in Eastern Serbia. In: Rapantova, N., Hrkal, Z. (Eds.), *Mine Water and the Environment*. Paper 67, Ostrava (VSB – Technical University of Ostrava).
- IAEA (International Atomic Energy Agency), 2003. Guidelines for radioelement mapping using gamma ray spectrometry data. Vienna, Austria.
- Iyengar, M.A.R., 1990. The natural distribution of radium, The Environmental Behaviour of Radium, Vienna: International Atomic Energy Agency. Technical Report Series No. 310, Vol. 1. Vienna: International Atomic Energy Agency, pp. 59–128.
- Factor analysis using SAS PROC FACTOR, 1995. Online: <http://www.cc.utexas.edu/cc/docs/stat53.html>.
- Gros, N., 2003. The comparison between Slovene and central European mineral and thermal waters. *Acta Chim. Slov.* 50, 57–66.
- Hopke, P., 1985. *Receptor Modeling in Environmental Chemistry*. Wiley, New York.
- Jackson, D.J., 1993. Stopping rules in principal components analysis: a comparison of heuristic and statistical approaches. *Ecology* 76, 2204–2214.
- Jang, C.S., 2010. Applying scores of multivariate statistical analyses to characterize relationships between hydrochemical properties and geological origins of springs in Taiwan. *Journal of Geochemical Exploration* 105, 11–18.
- Jia, G., Torri, G., 2007. Estimation of radiation doses to members of the public in Italy from intakes of some important naturally occurring radionuclides (²³⁸U, ²³⁴U, ²³⁵U, ²²⁶Ra, ²²⁸Ra, ²²⁴Ra and ²¹⁰Po) in drinking water. *Applied Radiation and Isotopes* 65, 849–857.
- Kaiser, H.F., Rice, J., 1974. Little Jiffy, Mark IV. *Educational and Psychological Measurement* 34, 111–117.

- Kitto, M.E., Parekh, P.P., Torres, M.A., Schneider, D., 2005. Radionuclide and chemical concentrations in mineral waters at Saratoga Springs, New York. *Journal of Environmental Radioactivity* 80, 327–339.
- Liu, C.W., Lin, K.H., Kuo, Y.M., 2003. Application of factor analysis in the assessment of groundwater quality in a blackfoot disease area in Taiwan. *The Science of the Total Environment* 313, 77–89.
- Milivojević, M., 1989. Assessment of Geothermal Resources of Serbia Excluding Autonomous Provinces (Doctor thesis), University of Belgrade, Belgrade (in Serbian).
- Milivojević, M., 1991. Elaborate on the Exploitation Potentials of Thermal and Mineral Water of the Niška Banja Springs. In: *Laboratory for Geothermal Energy* (Ed.), University of Belgrade, Faculty of Mining and Geology, Group for Hydrogeology, Belgrade, pp. 29–32 (in Serbian).
- Milivojević, M., 2003. Carbogaseous mineral water in Serbia and BiH as indicator of deep hydrogeothermal resources. *European Geothermal Conference*, Szeged, Hungary, pp. 26–29. May 2003, P-5-03.
- Miljević, N.R., Golobčanin, D.D., Milivojević, M.M., 1996. A stable isotope study of geothermal waters in Serbia. *J. Serb. Chem. Soc.* 61, 831–840.
- Murtagh, F., Heck, A., 1987. *Multivariate Data Analysis*. Kluwer, Dordrecht. <http://strule.cs.qub.ac.uk/~fmurtagh/books/holdall.pdf>.
- Nikic, Z., Kovacevic, J., Papic, P., 2008. Uranium in the Groundwater of Permo-Triassic Aquifers of the Visok Region, Stara Planina, Eastern Serbia. *Water, Air, and Soil Pollution* 192, 47–58.
- Petrović, T., Zlokolica-Mandić, M., Veljković, N., Vidojević, D., 2010. Hydrogeological conditions for the forming and quality of mineral waters in Serbia. *Journal of Geochemical Exploration*. doi:10.1016/j.gexplo.2010.07.009.
- Piper, A.M., 1944. A graphic procedure in the geo-chemical interpretation of water analyses. *Trans Am Geophys Union* 25, 914–923.
- Protić, D., 1995. Mineral and thermal waters in Serbia, Book 17. Geoinstitut, Belgrade. (in Serbian).
- Ramasamy, V., Suresh, G., Meenakshisundaram, V., Ponnusamy, V., 2011. Horizontal and vertical characterization of radionuclides and minerals in river sediments. *Applied Radiation and Isotopes* 69, 184–195.
- Shapiro, S.S., Wilk, M.B., 1965. An analysis of variance test for normality (complete samples). *Biometrika* 52, 591–611.
- Sneath, P.H.A., Sokal, R.R., 1973. *Numerical Taxonomy*, San Francisco, W. H. Freeman. pp. 295–296.
- Tanasković, I., Golobčanin, D., Komatina Petrović, S., Miljević, N., 2011. Natural radioactivity in Serbian spa waters. *Journal of Environmental Protection and Ecology* 12, 286–294.
- van Wirdum, G., 1980. Description of water-quality changes in hydrological cycle, for the purpose of nature conservation. Reports and notes In: Hooghart, J.C. (Ed.), *Water quality in groundwater-flow systems*, 5. Commission for Hydrological Research TNO, The Hague, The Netherlands, pp. 118–143 (in Dutch).
- Wenning, R.J., Erickson, G.A., 1994. Interpretation and analysis of complex environmental data using chemometric methods. *Trends in Analytical Chemistry* 13, 446–457.
- Yu, J.C., Quinn, J.T., Dufournaud, C.M., Harrington, J.J., Roger, P.P., Lohani, B.N., 1998. Effective dimensionality of environmental indicators: a principal component analysis with bootstrap confidence intervals. *Journal of Environmental Management* 53, 101–119.
- Žunić, Z.S., Kobal, I., Vaupotić, J., Kozak, K., Mazur, J., Birovljev, A., Janik, M., Čeliković, I., Ujić, P., Demajo, A., Krstić, G., Jakupi, B., Quarto, M., Bochicchio, F., 2006. High natural radiation exposure in radon spa areas: a detailed field investigation in Niška Banja (Balkan region). *Journal of Environmental Radioactivity* 89, 249–260.